Problem 1

TISP variable selection method With threshold operator:

$$\Theta(x, \lambda, \eta) = \begin{cases} 0 & \text{if} |x| \leq \lambda \\ \frac{x}{1+\eta} & |x| > \lambda \end{cases}$$

Penalized logistic regression:

$$L(\boldsymbol{\omega}) = \frac{1}{N} \sum_{j=1}^{N} \ln(1 + e^{-y_j^* \boldsymbol{\omega}^T \boldsymbol{x}_j}) + \lambda \sum_{i=1}^{p} p(\omega_i)$$

assuming $y_j^* \in \{-1, 1\}$ and algorithm iteration:

$$\boldsymbol{\omega}^{t+1} = \Theta \left(\boldsymbol{\omega}^t + \eta' X^T \left[\boldsymbol{y} - \frac{1}{1 + e^{-X \boldsymbol{\omega}^t}} \right], \lambda \right)$$

assuming $y_j^* \in \{-0, 1\}$